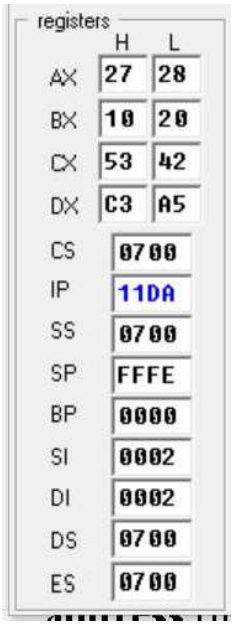
# Lab 02
# Variables and Memory in 8086 Assembly

**Object:**

- Define the variable in 8086 assembly language.
- Find these variables in memory.
- Check the size of variables and the related registers.
- Transfer a values from variables into the registers

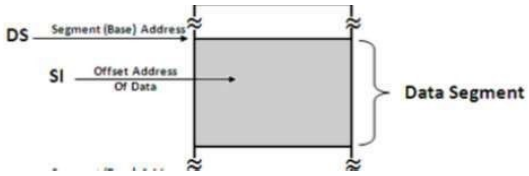





# Memory Segment

- Therefore, the **physical address** (**20-bit**) could be calculated using the following equation:

$$Physical_{Address} = [Segment_{Address}] \times 10]_H + \text{Offset}_{Address}$$




Only DS is required for this lab ➔

| Segment Register | Offset Register |
|---|---|
| DS | SI, DI, BX |

**Note**:
Also, the offset could be assigned through variable name. The CPU will convert the variable name into an offset number in Hex!

3

# Variables

Variable is a memory location. Our compiler supports two types of variables:

BYTE and WORD.

| Syntax for a variable declaration: | |
|---|---|
| name DB value | DB - stays for Define Byte. |
| name DW value | DW - stays for Define Word. |

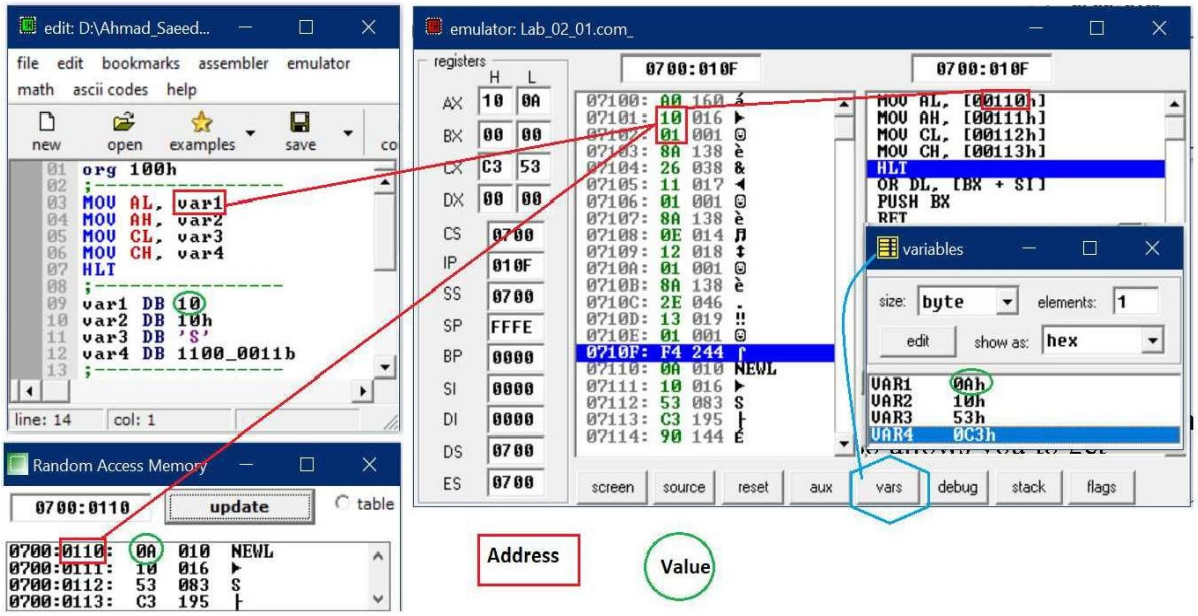
**Name**:
- can be any letter or digit combination
- it should start with a letter.

**Value:**
- can be any numeric value (hexadecimal, binary, or decimal).
- "?" symbol for variables that are not initialized.

4

# Example 1

Load the registers, AL, AH, CL, and CH with the variables var1, var2, var3, and var 4, respectively. Also, set these variables with 10, 10h, 'S', and 1100_0011b.




Try to load (10) into the register AX, what is the message your got? Why?!

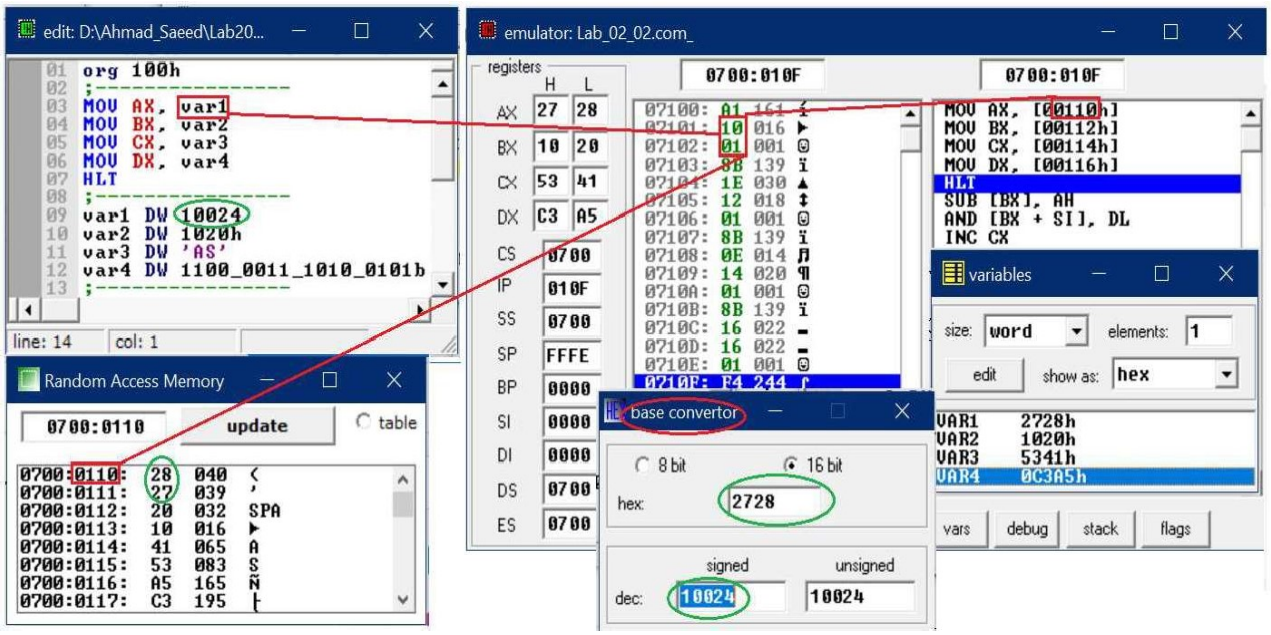
Why the (10) in code line (8) Appear as (0A) in register AL?

Why the ('S') in code line (10) Appear as (53) in register CL?



# Example 2

Load the registers, AX, BX, CX and DX with the variables var1, var2, var3, and var 4, respectively. Also, set these variables with equivalent in Dec of (2728h), 1020h, 'AS', and 1100_0011_1010_0101b.




Try to load 2020 into the register AL, what is the message your got?

What is the memory location of var3?

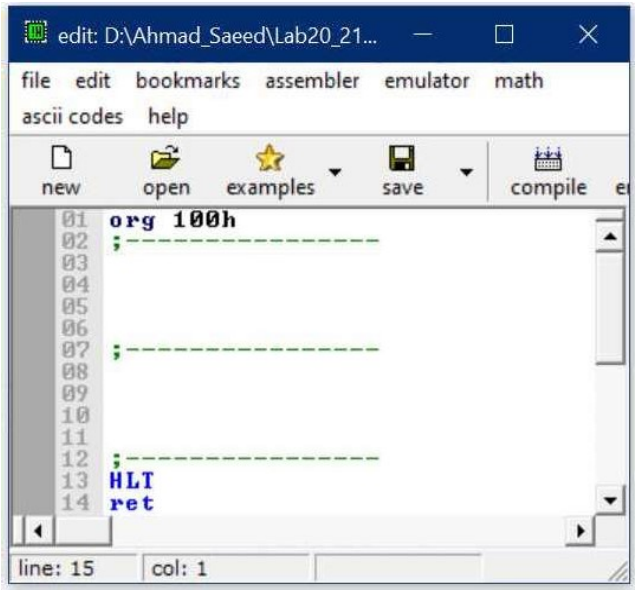
# Example 3

Load the registers as the following (In Lab!):

| Register | Variable Name | Description |
|---|---|---|
| AX | Year | Birth year of your father in Hex! |
| Bh | Age | The age of your big brother or sister in Dec.! |
| CX | Initials | First letter of your name and first letter or your father name. |
| DX | Date | YY,DD of your birthday. For example: (07-15) is 15th of July! Each digit in binary. |

**Questions**:

- What is the memory location of Age and Date?
- What is the values of Dh and DL?
- What is the value of Bh?
- How many bytes does Year took? Try to show the window variables from (vars) button in the bottom of emulator!






# Homework

Write an assembly code to do the following:

1. Set the register AL and DH with immediate (direct) values (10h) and ('T') as your learn from the previous lab!
2. Then, transfer the content of AL and DH into the memory location Loc1 and Loc2.
3. Write down the base address and the effective address (offset).

